[Total No. of Questions: 09]

Morning

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Uni. Roll No. ................

Program/ Course: B.Tech. (Sem- 4th)
Name of Subject: Computer Architecture and Microprocessors
Subject Code: IT-14405
Paper ID: 15336

**Time Allowed: 3 Hours** **Max. Marks: 60**

**NOTE:**

1) **Section-A** is **compulsory**
2) Attempt any **four** questions from **Section-B** and any **two** questions from **Section-C**
3) Any missing data may be assumed appropriately

## Section – A

**[Marks: 02 each]**

**Q1.**

a) How many memory chips of 256 × 8 are needed to provide a memory capacity of 4K bytes? And how many address lines are required to access 4K bytes of memory?
b) Differentiate BUN and BSA instructions along with suitable example
c) Define the following: Micro-instruction and Micro-program
d) Define RIM and SIM Instructions
e) Discuss the role of ALE in 8085 Microprocessor
f) Define hardware and software interrupts. Name the Interrupt having highest priority
g) List few applications of Microprocessor based Systems.
h) Write 1-address and zero address instruction for expression: (A+B)*(C+D) .
i) Differentiate 8085 and 8086 Microprocessor Architecture.
j) Define Virtual Memory and its need.

## Section – B

**[Marks: 05 each]**

**Q2.** A digital computer has a memory unit of 64K × 16 and a cache memory of 1K words. The cache uses direct mapping with a block size of four words. How many bits are there in the tag, index, block and word fields of the address format?

**Q3.** Define Interrupt. Draw Flowchart for Interrupt Cycle and explain how to handle interrupt

**Q4.** Give the Comparison between Hardwired Control and Micro-programmed Control Organization

**Q5.** Define Assembly Language Programming. Explain Data Transfer Operations

**Q6.** Explain in detail the difference between RISC and CISC Architecture

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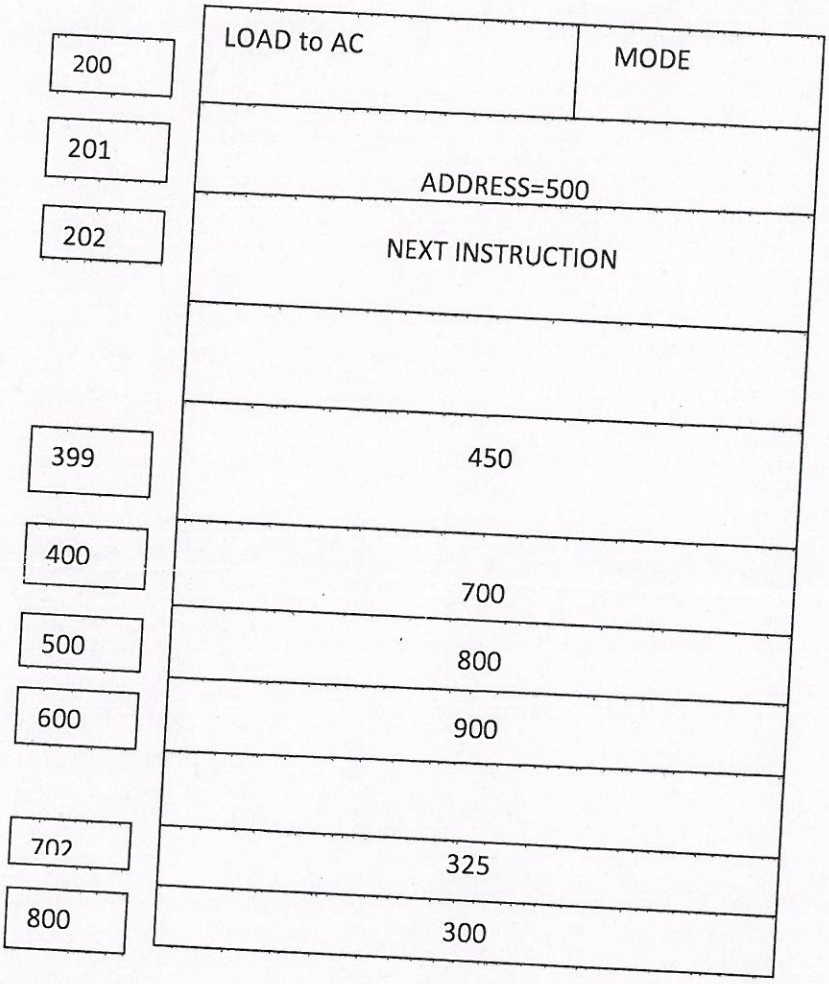
## Section – C [Marks: 10 each (05 for each sub-part, if any)]

**Q7.**. What do you mean by addressing modes? If there is a two-word instruction as shown in diagram and value of PC=200, Register value=400, index register =200.Then using all addressing modes: Direct Address, Immediate Address, Indirect Mode , Relative Mode, Index Mode, Register Mode and Register Indirect Mode, find the value of EFFECTIVE ADDRESS and ACCUMULATOR in each of the addressing modes




**Q8.** Define Microprocessor. Explain the 8085 Microprocessor Architecture in detail

**Q9**. What do you mean by Initialization of DMA Controller? How DMA Controller works? Explain with suitable diagram

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